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Companion to *A Topological Tidal-Curvature Imager* — Draft for comment

A Placement Law for Shift Sensing in Curved-Spacetime Su-Schrieffer-Heeger Chains

Draft for comment — not peer-reviewed. Companion to *A Topological Tidal-Curvature Imager* (tidal-imager-draft), whose curvature placement law this result extends to stationary metrics. Every claim is DERIVED, COMPUTED, or absent.

Authorship note. This work was developed in extended collaboration between B. Watson (direction, framing, verification protocol) and AI systems (Claude, Anthropic; two independent sessions), which performed the formal derivation, numerics, and drafting under the author's direction. The two-session cross-verification protocol is described in Appendix B and is, to our knowledge, itself a methodological novelty.

Abstract

The curved-spacetime Su-Schrieffer-Heeger (SSH) chain encodes a static diagonal metric in its bond profile and reads it through a topological edge mode; its curvature sensitivity obeys a placement law derived in companion work. Stationary metrics — the Alcubierre class — additionally carry a shift vector g_tx, and the lattice term that encodes a shift is a current-like same-sublattice hopping, which breaks the chiral symmetry protecting the readout. We show this costs nothing where it matters: an antiunitary chiral symmetry (sublattice parity composed with complex conjugation) survives for arbitrary shift profiles, pinning the edge mode at zero energy to machine precision even under bond disorder, while the naive real form of the same coupling unpins the mode at first order. We then derive the shift placement law: the logarithmic response of the edge density of states equals minus the leaked probability, $d \ln \rho = -\tilde{w}^2 |\psi_0|^2$, where $\tilde{w}$ is the shift in local-hopping units and $|\psi_0|^2$ the mode's probability weight at the shifted bond. The derivation shows the entire disturbance exits through exactly one site of the unprotected sublattice, making the response kernel exactly diagonal at leading order. Corollaries: the observable is exactly even in the shift — static calorimetry reads $|g_{tx}|^2$ and is structurally blind to its sign; curvature bias preamplifies shift response (computed 70–85× for the profile studied, the spread being hopping-convention dependent); and a synthetic warp wall of two-percent shift amplitude parked on one element is detectable at SNR ≈ 16–20 in 100 s with demonstrated cryogenic bolometry. An array separates the curvature and shift channels if and only if neighboring elements differ in curvature bias (condition number ≈ 5); shift-bias diversity does not work. All results verified by two independent implementations in different conventions.

1. Introduction

The companion white paper develops a curvature imager: a dimerized chain whose position-dependent hopping encodes a static diagonal metric, probed by a weakly coupled qubit whose decoherence power reports the edge mode's local density of states, governed by a placement law — sensitivity equals the differential metric tilt integrated over the mode's probability footprint. That instrument's encoding dictionary, inherited from the curved-spacetime SSH construction of Rajbongshi and Modak, covers static diagonal metrics only. The metrics of principal interest to the analog-warp community are stationary: their content lives in a shift vector g_tx, and a sensing program aimed at engineered warp-class geometries must answer a prior question — can a shift be written into the lattice at all without destroying the topological protection the readout is built on?

The concern is concrete. The lattice route to a tilted cone — the dispersion signature of a shift — runs through same-sublattice hopping, and any same-sublattice term breaks the chiral symmetry that pins the edge mode at zero energy and gives the calorimetric

readout its contrast. If protection and encodability draw from the same account, the shift-sensing program is structurally self-defeating. This paper computes and then derives the answer: they do not. The correct encoding preserves a weaker antiunitary protection that does the same pinning work; the mode's shape, unlike its energy, responds to the shift; and that response obeys a placement law as clean as the curvature one, with a one-line closed form. The instrument class that emerges is a magnitude detector for synthetic frame-dragging, with its blind spots derived rather than discovered.

2. Encoding a shift, and what protects the mode

Two same-sublattice couplings present themselves. The real next-nearest-neighbor hopping t' contributes a k-even identity term to the dispersion — an energy offset, not a tilt — and it is a protection catastrophe: computed on the warped chain, the zero mode's energy shifts linearly, $E_0 = -0.73\, t'$ (in the $t_1 t_2 = 1$ convention), confirmed over four decades. The current-like imaginary coupling, $iw(c^\dagger_{n+2} c_n - \text{h.c.})$, contributes $2w \sin k$ — the odd-in-k identity term that tilts the cone and is the lattice form of the shift. Computed on the same chain: the zero mode stays at machine zero ($|E_0| \le 10^{-16}$) to $w = 0.1$, including under one-percent bond disorder across 200 realizations, where the real form at $t' = 0.02$ is unpinned to $|E_0| \approx 1.4 \times 10^{-2}$.

The mechanism is a symmetry handover. The unitary chiral operator S (sublattice parity) no longer anticommutes with H once any same-sublattice term is present. But for the current-like term, the antiunitary operator $S \circ K$ does: complex conjugation K flips the sign of the imaginary coupling while S flips the sign of the real nearest-neighbor bonds, so $(S \circ K) H (S \circ K)^{-1} = -H$ exactly, for any shift profile $w(x)$ — the argument never references the profile. The spectrum remains rigorously $\pm E$ paired (verified below 10^{-14} at $w = 0.1$ with disorder, i.e. at the floor of double precision for a 201-site matrix), and the unpaired mode of an odd chain is pinned at zero. The vulnerability ledger is unchanged by the shift term: on-site potentials and real same-sublattice hopping remain the enemies of the protection; the shift adds no new class.

3. The shift placement law

The pinned mode reshapes under the shift, and the edge observable tracks the reshaping. Write the chiral chain in sublattice blocks, $H_0 = [[0, h],[h^\dagger, 0]]$, with the zero mode $|0\rangle = (a, 0)$ real, normalized, purely on sublattice A, obeying the product structure $a_{m+1}/a_m = -t_1(m)/t_2(m)$. Perturb with one shift bond of strength w on the A–A link between cells m and m+1. Four steps, each verified numerically to the precision of the method:

First, the first-order correction leaks entirely into sublattice B: the perturbation applied to $|0\rangle$ is A-supported with vanishing diagonal element, and the chiral pairing of the excited states cancels its A-components between $\pm E$ partners while their B-components add. The edge site is an A site, so the reading is untouched at first order — the evenness of the observable in w is derived here, not assumed.

Second, the energy stays pinned: the first-order shift vanishes by antisymmetry and the second-order shift vanishes because the perturbation is A-only while the first-order correction is B-only. This is the perturbative face of the $S \circ K$ pinning.

Third — the structural surprise — the leak occupies exactly one site. Solving $h\, \psi_B = -(V|0\rangle)_A$ site by site from the closed end, every B amplitude vanishes until cell m, where $\psi_B(m) = iw\, a_{m+1}/t_1(m)$; at cell m+1 the two contributions cancel *identically by the zero-mode ratio*, and everything beyond stays zero. The same product structure that builds the mode confines its leak. Verified: one spike, predicted site, predicted amplitude to five significant figures, all other B amplitudes at 10^{-16}.

Fourth, the second-order change to the A-sublattice shape is a pure renormalization — the perturbation annihilates the B-only first-order correction, so the $O(w^2)$ term is proportional to the zero mode itself, fixed by normalization. The edge reading therefore drops by exactly the leaked probability:

$$\mathbf{d \ln \rho_{edge} = -\|\psi^{(1)}\|^2 = -w^2 \, a_{m+1}^2/t_1(m)^2 = -w^2 \, a_m^2/t_2(m)^2}$$

The two faces are equal by the mode ratio; in the canonical dimensionless variable $\tilde{w} = w/t_2(m)$ the law is pure footprint, $d \ln \rho = -\tilde{w}^2 \, a_m^2$. In words: **the shift sensitivity of the edge reading equals the squared local shift-to-hopping ratio weighted by the mode's own probability footprint — sensitivity is leaked probability.** This is the exact stationary-channel analog of the curvature placement law: same footprint structure, quadratic rather than linear, with the local ratio replacing the accumulated tilt.

4. Corollaries and channel structure

Sign blindness is structural. Complex conjugation maps H(w) to H(−w) with an identical spectrum, so every odd-order response vanishes identically: static calorimetry measures $|g_{tx}|^2$, the magnitude of the frame-dragging, and cannot measure its direction. Reading the sign requires a time-odd observable — directional transport — which is a different instrument, not a refinement of this one.

The response kernel is exactly diagonal at leading order, because each bond's leak occupies a distinct B site; measured off-diagonal support is confined to an immediate-neighbor cross term entering beyond leading order. B–B shift bonds do not couple to the zero mode at second order at all (the mode has no B weight for them to act on); the small quadratic residual measured on B bonds was traced to the non-topological bulk background transmitted through the Lorentzian linewidth, identified by its η^2 scaling (constant to four digits across a 30× range of η, with the A-bond response η-independent as a zero-mode property must be). It is an artifact channel nine orders below signal at operating parameters and shrinks as the engineered linewidth narrows.

Curvature bias is a shift preamplifier. The $1/t_2^2$ gauge factor in the law grows wherever the bond profile is soft; on the power-law warped chain the identical near-edge shift feature produces a 70–85× larger response than on the flat chain — 85× in the $t_1 t_2 = 1$ convention, 72× in the $t_2 = 1$ convention; the raw factor is convention-dependent because the shift amplitude is held fixed in absolute rather than local-hopping units, and the law makes the factor calculable for any profile and convention. A shift imager therefore wants a deliberately curved substrate.

And blindness beyond the footprint is the law's locality made literal: computed response to a warp bubble versus its position falls from 2×10^{-4} at five cells to exactly machine zero by thirty cells. The pixel is the footprint; a mid-chain feature is invisible to an edge element, and imaging extended shift structure requires the array.

5. Demonstration: a synthetic warp wall, with cited hardware

A 1+1 Alcubierre-style top-hat shift profile with smooth walls — flat inside, flat outside, all structure in two opposite-signed gradient walls — costs the mode nothing ($|E_0| \leq 10^{-18}$ with the bubble present) and produces a computed edge response of 1.4–1.8% in log-density for a two-percent shift amplitude parked near the element (the range reflects wall-geometry choices; both implementations agree within it). Against the companion paper's worked superconducting element — 3.3 aW signal power, demonstrated 30 zW/√Hz bolometry — that is 47–60 zW: **SNR ≈ 16–20 at 100 seconds of integration.** A static analog warp wall at two-percent shift amplitude is comfortably measurable with hardware already cited to demonstrated devices.

6. Array design rule: curvature-bias diversity

A single edge observable entangles the curvature and shift channels — the cross-responsivity $d \ln \rho/d\sigma$ itself runs with w. Computed response vectors at multiple element biases settle the disentangling scheme, and the convention-invariant statement is angular: a flat element paired with a warped element yields normalized response vectors separated by tens of degrees (38° in one implementation, comparable in the other), inverting the (σ, w^2) system stably; two elements differing only in shift bias are

collinear to a fraction of a degree and separate nothing — obvious in hindsight from the law, since both share the same $1/t_2^2$ soft-region weighting. Both independent implementations reach the same qualitative verdict; raw condition numbers are deliberately not quoted, having been found normalization-dependent between conventions (a hazard the canonical variable $\tilde{w}$ removes if adopted end-to-end). **The array separates curvature from shift if and only if neighboring elements differ in curvature bias.** Fabrication cost: one lithographic profile choice per element, no new hardware.

7. Validation path

The photonic stage of the companion paper's ladder carries this result at no extra cost beyond mask design: a warped dimerized waveguide array with a current-like coupling (realizable as directional/auxiliary-ring coupling) exposes three quantitative predictions at room temperature — the quadratic responsivity in w with its closed-form coefficient, the exact evenness under sign reversal of the shift, and the single-site structure of the leaked amplitude, which is directly imageable in a photonic lattice. A null on any of the three falsifies the law before cryogenic investment. The superconducting stage inherits the companion paper's single differential measurement with one addition: a shift-biased element beside an unbiased one tests the preamplification factor and the disentangling condition number in the same cooldown.

8. Honest scope

The dictionary is the standing debt. The identification of the current-like coupling with g_tx is by dispersion form — the odd-in-k identity term is the tilt of a 1+1 stationary metric — not by a completed static-to-stationary extension of the curved-spacetime SSH derivation. That formal task remains open, and this paper's recommendation to it is concrete: define the lattice shift in local-hopping units from the start, since the convention-dependence of every raw-w statement was measured directly when two independent implementations disagreed by exactly the gauge factor the law prescribes. All results here are static; the response of the protected mode to a moving wall — formation and collapse analogs, the frontier the gravitational-waveform community has named — is untouched by these methods. Sign readout is excluded by symmetry, not difficulty, and belongs to a transport architecture. Array conclusions are two-element condition-number statements; full-array error propagation under fabrication scatter is not yet computed. And the entire result chain, while doubly implemented and cross-verified, has not been examined by anyone outside the two sessions that produced it.

9. Conclusion

The shift can be written into a topological chain without paying for it in protection: the symmetry changes hands rather than disappearing, the mode's energy stays pinned for arbitrary shift profiles, and the mode's shape becomes the sensor. Its law is one line — sensitivity is leaked probability, footprint-weighted — and it hands the analog-gravity community the same offer as the curvature law, extended to the metric class they actually care about: send the stationary metric, the law returns its lattice image and the response of every element placement, and a photonic array can falsify the whole construction for the cost of a mask.

Appendix A: Methods

Odd chains, 201 sites (100 cells), exact diagonalization and direct solve; power-law profile of the companion paper throughout; broadening $\eta = 10^{-3}$ as the engineered-linewidth proxy except in the η-scaling study (3×10^{-3} to 10^{-4}). Conventions: session one, $t_1 t_2 = 1$; session two, $t_2 = 1$; all convention-covariant claims verified in both. Scripts (persisted with this draft): tilt_cost.py, tilt_cost2.py, bubble.py, reciprocal.py, derive_check.py (session one); verify_return.py, kernel_law.py, law_check.py, derive_verify.py (session two).

Appendix B: Two-session verification protocol

Every result in this paper passed through two AI sessions with no shared code: claims were produced and computed in one session, reproduced from scratch in an independent implementation with a different hopping convention in the other, with pre-registered expectations where the protocol had matured. The record includes one productively falsified pre-registration (a two-point kernel conjecture, refuted by the exact diagonality later derived), one convention discrepancy opened in one session and closed in the other by the disputed law's own gauge factor, and one corollary error (a quartic-order attribution) caught by the counterpart session's verification and resolved by an η-scaling measurement that identified the true mechanism. Tears are recorded, not smoothed; they are the evidence the checking was real.

References

[1] Companion draft: *A Topological Tidal-Curvature Imager* (B. Watson, 2026, unpublished; carries the full citation list for the SSH model, the curved-spacetime construction, qubit-probe prior art, superconducting platform parameters, and bolometry). [2] A. Rajbongshi and R. Modak, arXiv:2504.16200 (2025). [3] Warp-metric physicality auditing: arXiv:2404.03095; Class. Quantum Grav. 41, 095009 (2024). *[verify before submission]* [4] Warp collapse waveforms: Clough, Dietrich, Khan, arXiv:2406.02466; Open J. Astrophys. 7 (2024). *[verify before submission]*